

Algebra 2

Unit 6: Solving Cube Root Equations — Review

Day 3: Practice & Review

Name: _____

Directions: Solve each equation. Show all work. Use the answer bank to check your answers. Not all answers will be used.

1. $x^{\frac{1}{3}} = 3$

5. $(2x - 14)^{\frac{1}{3}} = (x + 11)^{\frac{1}{3}}$

2. $(x + 1)^{\frac{1}{3}} - 2 = 3$

6. $3(x - 2)^{\frac{1}{3}} = 6$

3. $2\sqrt[3]{x} + 1 = 5$

7. $5 - x^{\frac{1}{3}} = 8$

4. $\sqrt[3]{2x + 5} = \sqrt[3]{4x - 9}$

8. $\sqrt{x} = -3$

Answer Bank

−27

−8

−3, 6

−2, 4

1

3

7

8

9

10

25

27

63

124

150

No solution

9. $3(7x + 1)^{\frac{1}{3}} = 4(3x)^{\frac{1}{3}}$

13. $\sqrt[3]{x^2 - 2x} = -1$

10. $(x^2 - 3x - 10)^{\frac{1}{3}} = 2$

14. The production output P of a factory is given by $P = 2\sqrt[3]{h - 25}$, where h is hours of labor. Find h when $P = 10$.

11. $\sqrt[3]{x^2 - 2x - 7} = 1$

15. $2\sqrt[3]{5x + 12} = 3\sqrt[3]{2x + 2}$

12. $(2x - 1)^{\frac{1}{3}} = 5$